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The Impact of Environmental Variability on Fishers' Harvest Decisions in Chile

A multi-species structural approach

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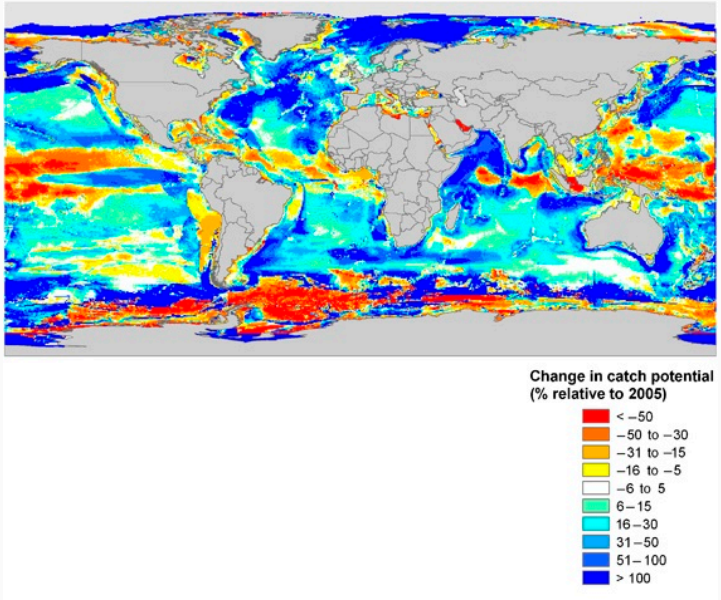
ICES/PICES Small Pelagic Symposium — La Paz, México, May 8, 2026

Motivation

- Small pelagic stocks fluctuate sharply — biology, prices, regulation, weather all move year-to-year.
- Standard projection tools assume **constant fishing rates or full quota uptake** — the choice is assumed away.
- But fishers **choose**: whether to participate, which species to target, how many trips to take.
- These choices are **mediated** by quotas, portfolio composition, and operating constraints.



Source: Cheung et al. (2010)

> ***My approach:*** *treat fisher participation as a count process — estimate it separately by fleet, then propagate to comparative statics.*

The project — two papers, one structural pipeline

Paper 1 (today's results) *Reduced-form effort + identified stock dynamics*

- **Negative binomial** trip equation, estimated separately by fleet.
- Bayesian state-space stock dynamics with structural climate shifters (in appendix).
- Comparative statics under counterfactual scenarios.

Paper 2 (today's roadmap) *Welfare counterfactuals*

- Translog **cost** function (trip-level, by fleet).
- **Inverse demand** (IAIDS, 3SLS).
- **Dynamic vessel optimization** $\Rightarrow$ optimal $T_g, h_{g\tau}$ under counterfactual shocks.
- Welfare, prices, **species substitution**.

> *The two papers share the **same identified stock dynamics**. Paper 1 stops at participation; Paper 2 closes the loop through prices, cost and profits to estimate optimal harvest and quota allocation.*

Setting and observed heterogeneity

Chile's Small Pelagic Fishery (Centro-Sur)

Stocks & sectors

- **Anchoveta** (*Engraulis ringens*), **sardine** (*Strangomera bentincki*), **jack mackerel** (*Trachurus murphyi*).
- ~94% of national landings; mostly purse seiners.
- Two sectors: **artisanal** (ART), **industrial** (IND).

Quota regime

- TAC split between sectors with **limited cross-sector transferability**.
- IND quotas partially tradable within sector; ART under spatial collective-property rights (i.e. **RAE**).

Fleet portfolios are very different

- **ART**: ~ {anchoveta, sardine}. Concentrated.
- **IND**: ~ 95% jack mackerel + 5% sardine.

Mobility

- IND: offshore purse-seiners.
- ART: port-anchored, RAE binds spatially.

$\Rightarrow$ Same shock, **different choice sets**.

Observed strategy switching — ART vs IND

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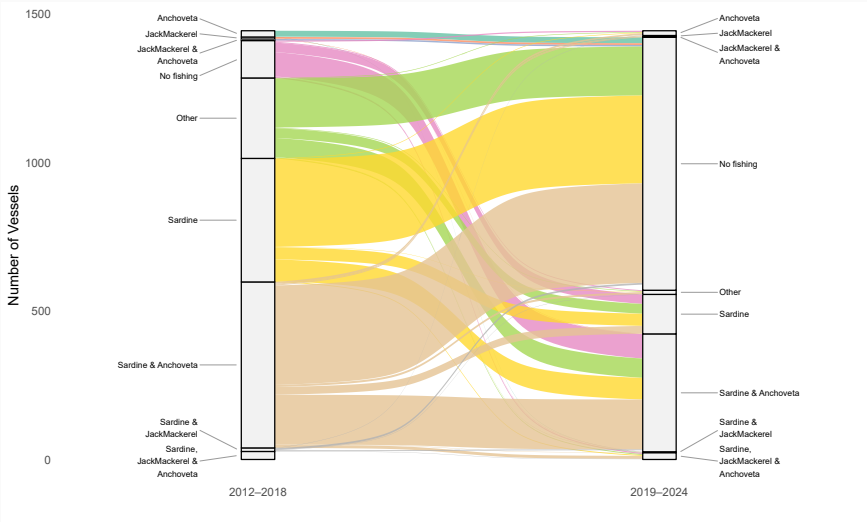
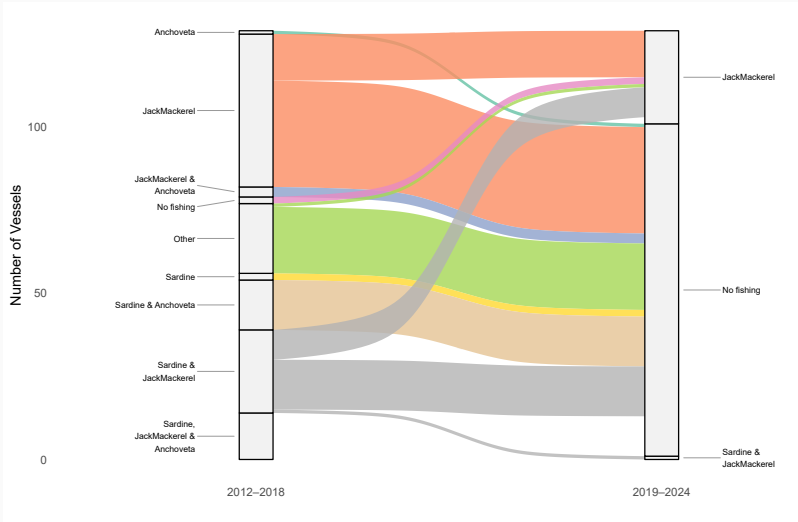
Artisanal



Industrial



Trip-level target-species transitions, IFOP logbooks 2013–2024.

> *ART: anchoveta↔sardine dominates; jack mackerel essentially absent. IND: rotation across all three. Same fishery, very different revealed choice sets.*

Paper 1 — The NB trip equation

A negative binomial model of annual trips

Specification, by fleet g (ART, IND)

$$T_{vy}^{g} \sim \mathrm{NB}\big(\exp(U_{vy}'\beta_g),\, \theta_g\big)$$

with $U_{vy} = \big[\, p_{sy},\ H_{vy}^{alloc},\ Z_v,\ O_{vy} \,\big]$:

- p_{sy}: ex-vessel prices, by species.
- H_{vy}^{alloc}: vessel-level quota allocation.
- Z_v: hold capacity, vessel type.
- O_{vy}: bad-weather days, closures.

Why NB, why by fleet?

- Overdispersion σ^2/μ: **22.4** (ART), **4.9** (IND). Poisson rejected at $p < 0.001$.
- Two fleets, two technologies, two choice sets — no pooling.
- Vessel-clustered robust SEs.

Identification of β_g

Quota shares vary across **vessels** $\times$ **years** (regional TAC moves; vessel share fixed). Reduced-form mapping.

What drives fisher participation?

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Variable	Industrial	Artisanal	Reads as
H_{vy}^{alloc}	**(+)** ***	**(+)** ***	Quota = binding budget constraint, both fleets.
Price — jack mackerel	**(+)** ***	0	IND oriented to jack mackerel; ART has no exposure.
Price — sardine	0	**(+)** ***	Sardine is ART's primary target.
Price — anchoveta	0	**(–)** ***	Simultaneity sign.
Hold capacity (log)	0	**(+)** ***	Heterogeneity is *within* the artisanal fleet.
Bad-weather days	0	**(–)** ***	ART vulnerable; IND insulated by vessel size.
Closure days	**(–)** ***	**(+)** ***†	IND: largest single coef. ART: zone FE collinear.

Negative binomial, vessel-clustered robust SEs. Sign and significance shown; full table in appendix. * $p < 0.01$. † zone-FE collinear (not causally interpretable).

> *Two fleets, **completely different participation models**.*

Comparative statics — the fleet asymmetry

Fleet	Counterfactual	$\%\Delta T$ **(marginal)**	$\Pr(\text{portfolio loss} > 50\%)$
Artisanal	mid-century, moderate (SSP2-4.5)	**–8.6%**	**0.95**
	end-century, severe (SSP5-8.5)	**–9.5%**	**0.99**
Industrial	mid-century, moderate (SSP2-4.5)	–0.9%	0.12
	end-century, severe (SSP5-8.5)	–0.9%	0.12

$\%\Delta T$: change in annual trips vs baseline; *marginal* over all posterior draws. Cross-median across a 6-model CMIP6 ensemble.

- ART: ~9× larger contraction than IND; near-certain portfolio loss.
- *Conditional* on no collapse: –2% ART vs –1% IND.

- $\Rightarrow$ Asymmetry runs through **probability of portfolio collapse**; quota cross-sector limits lock it in.
- IND's protection rests on jack mackerel (long-run shifter **n.i.**).

Paper 2 — closing the loop

Three structural extensions ⇒ welfare

What gets added on top of Paper 1

(i) Trip-level cost function — translog $C_{vg}(w, h, x, Z, Env)$, SUR by fleet. Recovers **input demands** (fuel, labor, ice) via Shephard's lemma and elasticities of input substitution under climate shocks.

(ii) Inverse demand (IAIDS) — $\ln p_{iy}$ as a function of all species' landings, instrumented by SST, CHL, fuel prices. Resolves the simultaneity that forced the anchoveta-price puzzle in Paper 1.

(iii) Vessel dynamic optimization — $\max_{T_g, h_{g\tau}} \sum_\tau \delta^\tau [P(h)\, h - C_g(h)]$ s.t. quota cap. Solved per fleet, posterior over $(T_g^*, h_{g\tau}^*)$ under each scenario.

Counterfactuals it enables

- **Equilibrium prices** under reduced supply.
- **Species substitution & optimal quota allocation** — vessel target choice and social planner's $\bar{Q}_s$ jointly determined.
- **Policy reforms** as adaptation:
  - Cross-sector quota transferability.
  - RAE redesign (multi-zone permits).
  - State-contingent TACs.

Full equations in appendix.

Takeaways

- **Paper 1**: a fleet-specific NB model of annual trips reveals **two completely different participation technologies** in the same fishery — ART and IND respond to different prices, different operating constraints, different scales.
- **Paper 1**: under counterfactual shocks, the ART:IND participation asymmetry is **~9:1**, running through the **probability of portfolio collapse**; quota cross-sector limits lock it in.
- **Paper 2**: closes the loop — **input demand, inverse demand, vessel × social-planner joint optimization** — to deliver welfare counterfactuals: equilibrium prices, species substitution & optimal quota allocation, and policy reforms (cross-sector transferability, RAE redesign, state-contingent TACs).
- **Joint message for S9**: adaptive capacity is not a property of fishers; it is a property of the **bundle (portfolio × permit × institution × market)** they sit in. Identifying adaptation requires modeling all four dimensions jointly — pooling, fixed prices, or aggregate effort all bias the answer.

¡Gracias! / Thank you

Questions?

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Funded by ANID-Chile, FONDECYT Iniciación N° 11250223.

Appendix

Appendix A · Stock dynamics — spec & priors

Process (Schaefer with shifter on r):

$$B_{i,t+1} = B_{i,t} + r_{i,t} B_{i,t}\left(1 - \frac{B_{i,t}}{K_i}\right) - C_{i,t} + \varepsilon_{i,t}$$

with $r_{i,t} = r_i^0 \exp(\rho_i^{SST} \Delta SST_{t-1} + \rho_i^{CHL} \Delta \log CHL_{t-1})$.

Observation (log-normal): $\log B_{i,t}^{obs} = \log B_{i,t} + u_{i,t}$.

Three nested specs: **ind**, **omega**, **full**.

Priors:

- $r_i^0, K_i, \sigma_{proc}, \sigma_{obs}$ from IFOP / SPRFMO single-species assessments.
- $\rho_i^{SST}, \rho_i^{CHL} \sim \mathcal{N}(\hat{\rho}_i^{stress}, 1)$.
- Ω: LKJ.

Estimation: Stan / HMC, 4 chains $\times$ 2000 post-warmup; $\hat{R} < 1.01$, $ESS > 400$.

Appendix B · Climate shifters

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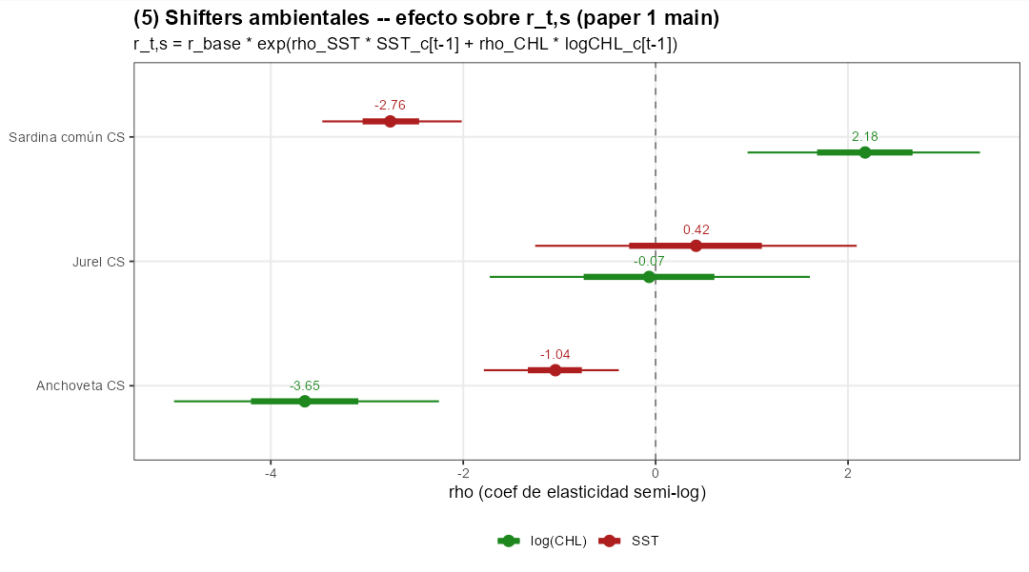


Posterior densities of ρ^{SST} and ρ^{CHL}, full multi-species specification. Anchoveta and common sardine: priors dominated by data. Jack mackerel: posteriors track priors $\rightarrow$ **n.i.** and reported as such throughout.

Appendix C · Stock productivity comparative statics

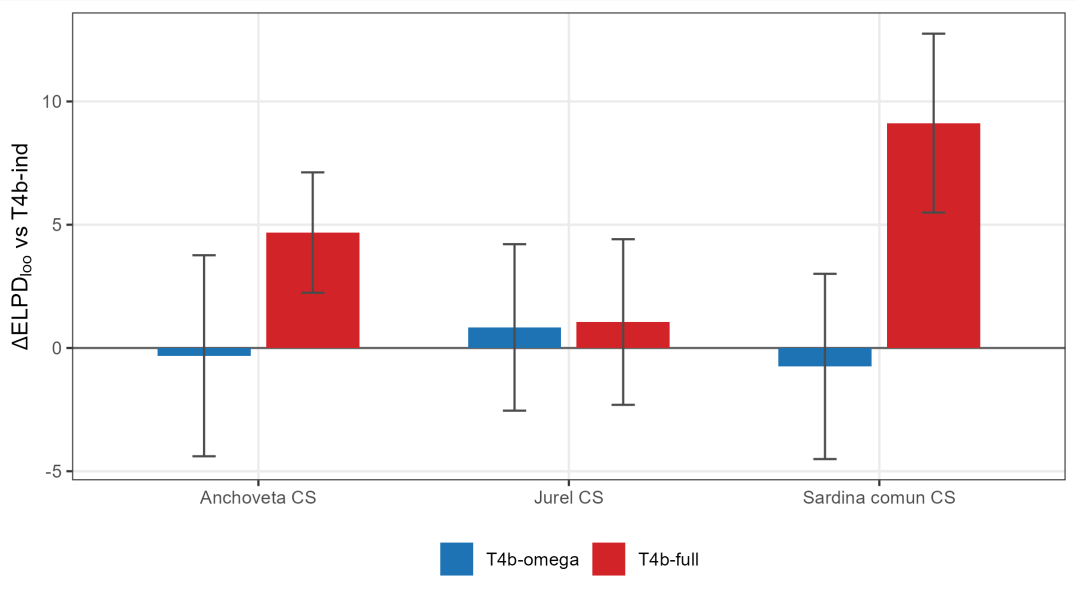
Stock	Scenario	$\Delta r/r_0$ (cross-median)	90% CI	$\Pr(\Delta < 0)$
Anchoveta	SSP2-4.5, mid-century	**–51%**	[–68%, –27%]	1.00
	SSP5-8.5, end-century	**–90%**	[–98%, –54%]	0.99
Sardine	SSP2-4.5, mid-century	**–79%**	[–86%, –67%]	1.00
	SSP5-8.5, end-century	**–100%**	[–100%, –99%]	1.00
Jack mackerel	all scenarios	n.i.	n.i.	n.i.

Cross-median across CMIP6 ensemble of 6 GCMs (CESM2, CNRM-ESM2-1, GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, UKESM1-0-LL); 90% CI is the within-posterior credible interval averaged across models.

Appendix D · Why is jack mackerel n.i.?

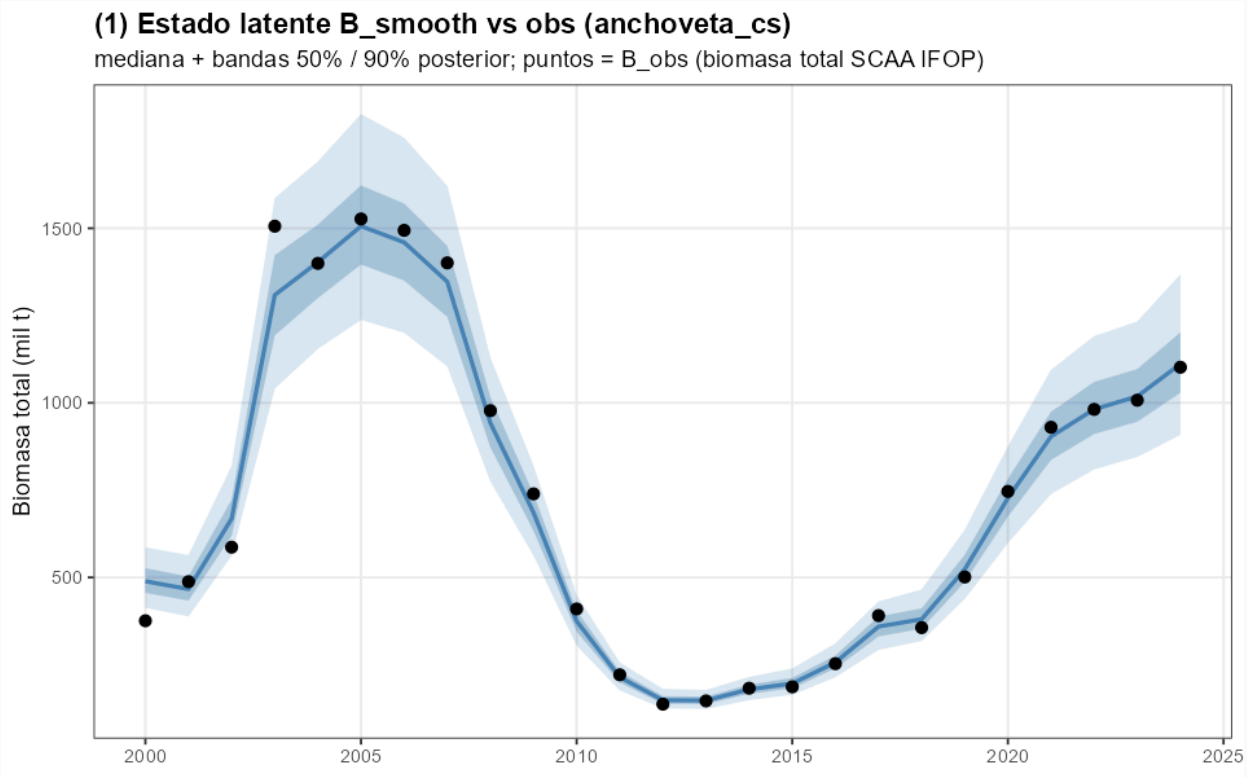
- Posteriors track priors across alternative spatial windows and biomass series.
- The CS biomass signal carries little information about climate shifters — pre-2002 acoustic gap, censored 2012/2015 observations, transboundary SPRFMO-scale dynamics.
- Reduced-form linear stress tests on growth increments do **not** close the hindcast under SST/CHL forcing alone, consistent with structural non-identification.
- **Implication:** climate-driven projection for jack mackerel CS must come from a **transboundary, game-theoretic** model (cf. Session S9, bullet 2), not a CS-only structural fit.

Appendix E · Model comparison (LOO)



LOO ELPD across specs: **ind** (Ω diagonal, $\rho = 0$), **omega** (Ω full, $\rho = 0$), **full** (Ω full, $\rho \neq 0$). Full wins at $\Delta\text{ELPD} \approx 14$–$18$; LFO one-step-ahead essentially tied.

Appendix F · Posterior predictive check



Smoothed posterior of biomass vs observed acoustic estimates, full model.

Appendix G · CMIP6 ensemble — ridgeline

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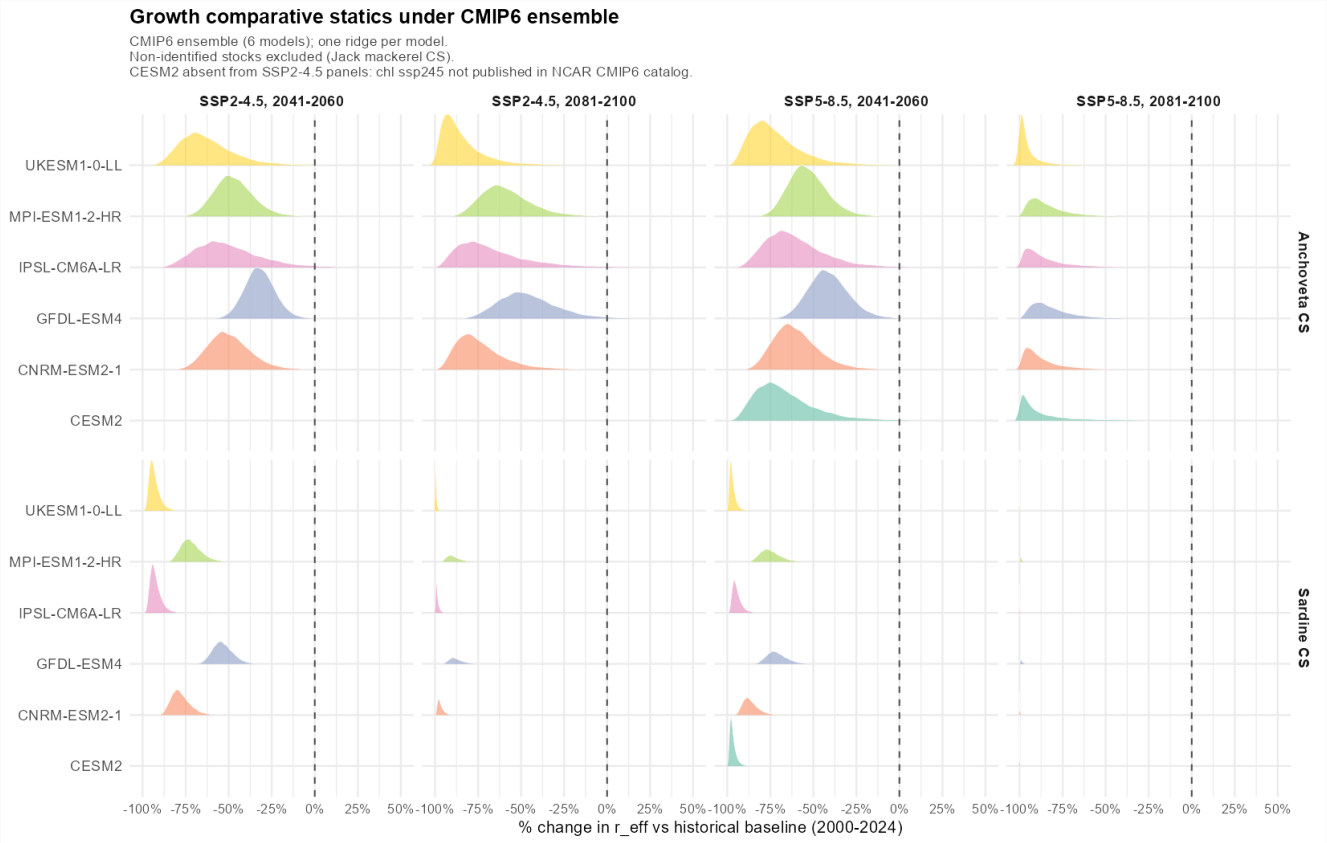

Posterior $\Delta r/r_0$ across SSP $\times$ period combinations, one ridge per CMIP6 model (6-GCM ensemble).

Appendix H · Variance decomposition of trip response

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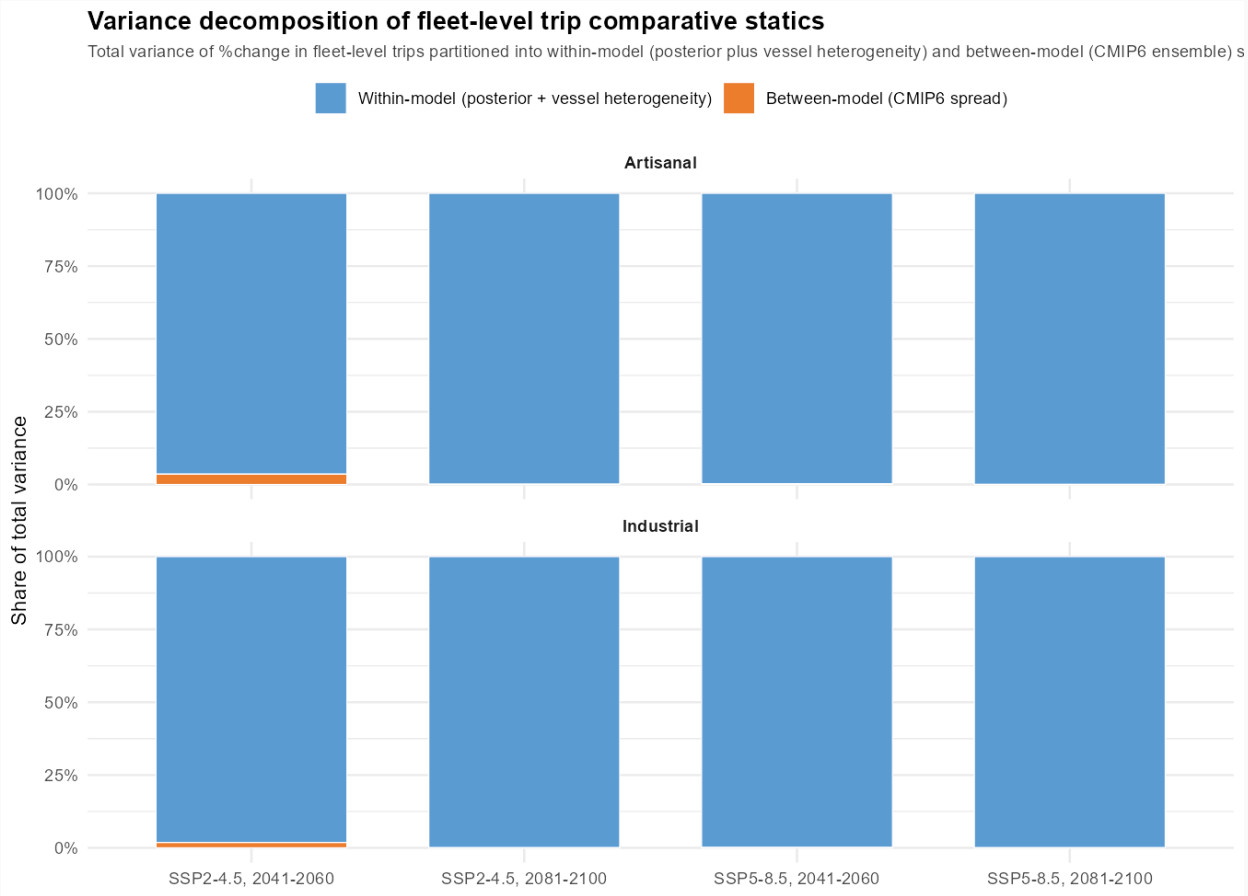

- **Within-model** (posterior + vessel heterogeneity) dominates: **96%** for ART SSP2-4.5 mid, saturating at **100%** under SSP5-8.5 end. IND: **98%** → **100%**.
- **Between-model** (CMIP6 ensemble): at most **4%**, often ≈ 0.
- The narrow cross-IQR in Table 5 is *not* climate-model consensus — it is a **floor**

Appendix I · Paper 2 — data inputs

Already in hand (Paper 1):

- Trip-level catch and price data (IFOP logbooks, 2013–2024).
- All-gear official catch series (SERNAPESCA, transparency request **AH010T0006857**, May 2025).
- Vessel characteristics & ports of operation.
- Environmental covariates (Copernicus, GlobColour, CMIP6 6-GCM ensemble).

Caveat — IFOP panel coverage (cross-validated vs SERNAPESCA vessel-level): IND 68% by vessel count, >90% by catch → representative; ART 32% by count, ~70% by catch → covers the **purse-seine artisanal** subset; non-PS gears (lampara, lines) excluded.

To request / acquire for Paper 2:

- Variable-input prices: crew wage, diesel cost, ice & supplies.
- Quota prices (auction or secondary markets, if available).
- Permits by vessel; quota holdings by sector.
- Fish-meal world price — demand instrument for IAIDS.